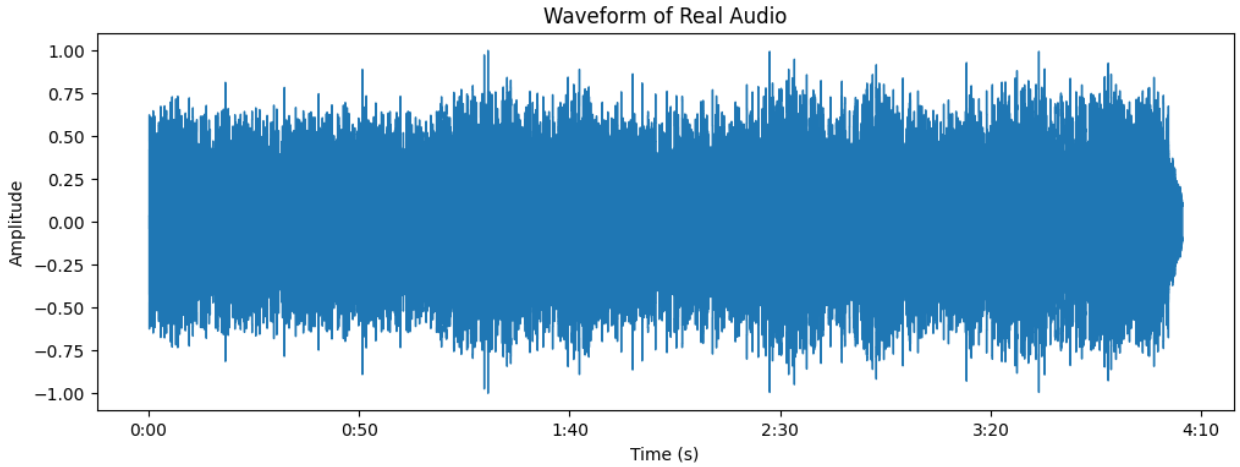
Hiring Assessment – AI Generated Songs detection

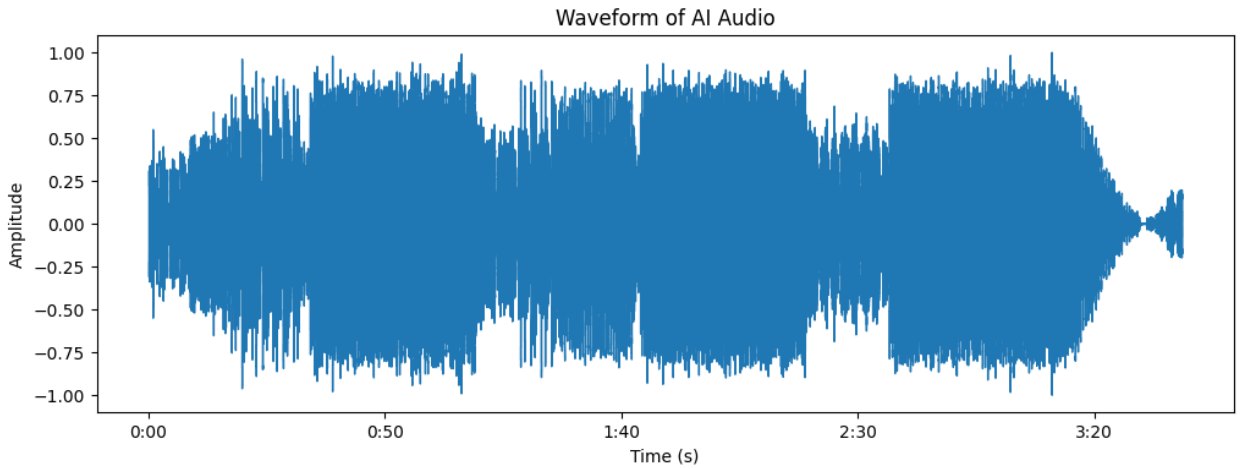
The objective of this assessment is to design and implement a robust pairwise similarity metric to detect and quantify the relationship between an original human-made track and a potential AI-generated counterpart. The focus is on relational modeling, determining if Track B is a synthetic derivative, variation or "hallucinated" generation based on Track A. Specifically, the goal is to develop a system that takes two full-length audio files as input and outputs a Similarity (Attribution) Score. This score must reflect the likelihood that the synthetic track is an attribution of the original.

Datasets

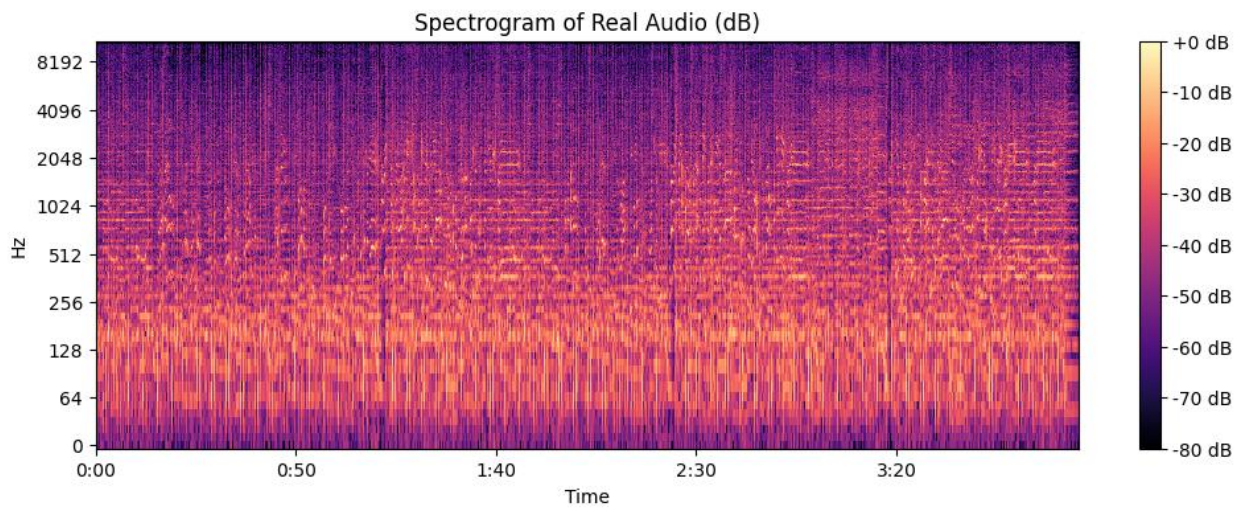
For this assignment, 3 datasets were proposed: SONICS, MIPPIA/SMP and FakeMusicCaps. FakeMusicCaps contains 10 second ai generated audio files from 5 generators. The MIPPIA/SMP dataset contains pairs of real songs that are related somehow and was the main dataset used for the analysis. SONICS contains a real song catalog and a fake song catalog. The fake songs are provided, while the real songs must be downloaded from YoutTube using yt-dlp library. In order to download them, the download.py script by MIPPIA/SMP was modified to get the youtube_id from the related csv file and download the audios in mp3 format. Due to YouTube anti-bot and rate limitations, this process needed adjustments in sleep seconds between calls and number of workers for parallelization. Additionally, to overcome failed calls caused by authentication requests, a cookies.txt file was passed as input to the yt-dlp run with extracted cookies from the browser. However, due to rate limitations the number of real songs that were downloaded, was significantly smaller than expected.

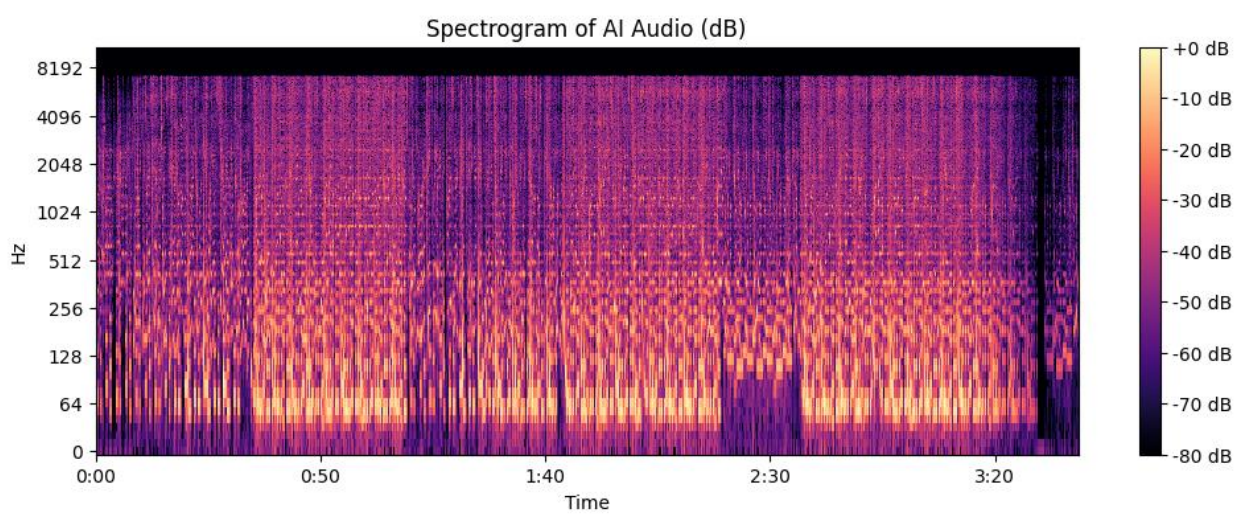
In the case of the SONICS dataset, the downloaded songs resulted in 3037 real and 49074 fake and in MIPPIA 10 pairs were dropped since only one of the songs was available for download. Below are some indicative waveforms of real and ai sogns.





Which makes us notice the more segmented patterns connected to ai generation.





These patterns are also visible in the Spectrogram visualization, where there are a lot of thin vertical lines across the entire frequency range, suggesting a lack of smoothed temporal evolution. Additionally, the changes focus mainly on the lower bands, compared to the real songs.

Exploratory feature extraction was conducted on both real and AI-generated music samples to understand their characteristics. Classical audio features, like MFCCs, chroma, spectral descriptors (centroid, rolloff, bandwidth), zero-crossing rate, and tempo were computed alongside higher-order features such as phase continuity, harmonic-to-noise ratio (HNR), and spectral flux. Initial observations suggested that timbral features (MFCCs) and harmonic descriptors (chroma) captured substantial variation in musical content, while certain signal-level features (HNR, phase continuity) showed limited discrimination between real and synthetic audio in this context. However, these are commonly used in AI audio detection since generated audio may exhibit unnatural tonal balance or inconsistent harmonic structures. However, because the task focuses on relational modeling, quantifying similarity between track pairs, rather than binary AI detection, all extracted features were retained for pairwise comparison. Additionally, DTW will be integrated to capture temporal correspondence.

Process

The problem is initially approached through a vanilla and interpretable baseline. Given two tracks, the system extracts descriptive features, computes multiple similarity measures and combines them into a final attribution score.

As the preprocessing step, the audio signals are resampled to a fixed sampling rate (22,050 Hz), converted to mono and optionally trimmed to remove silence. Each track is then divided into overlapping segments (currently defined as 15 seconds with 50% overlap) to capture temporal variability and reduce bias from any single portion of the audio.

Using the librosa library, several features are extracted from the segments. The features are split into classical and extended,

- Classical
    - MFCC to capture timbre
    - Chroma to capture harmonic and pitch content (key-invariant with 12-rotation matching)
    - Spectral features (centroid, rolloff, bandwith) for frequency distribution
    - Tempo, BPM
- Extended
    - Phase continuity, to capture phase artifacts
    - Harmonic to Noise Ratio
    - Spectral Flux, temporal spectral variation

Multiple similarity metrics are computed between the features of the two tracks, with mostly leveraging cosine similarity, normalized differences and DTW.

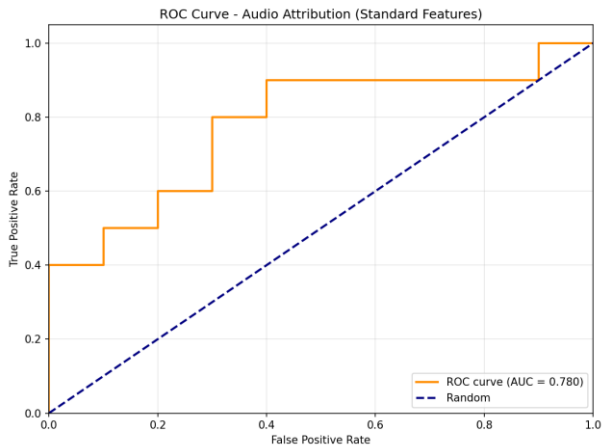
For the final score, a weighted combination of the individual similarity metrics is produced. Two configurations are used based on the standard or extended features used. The weights for the final

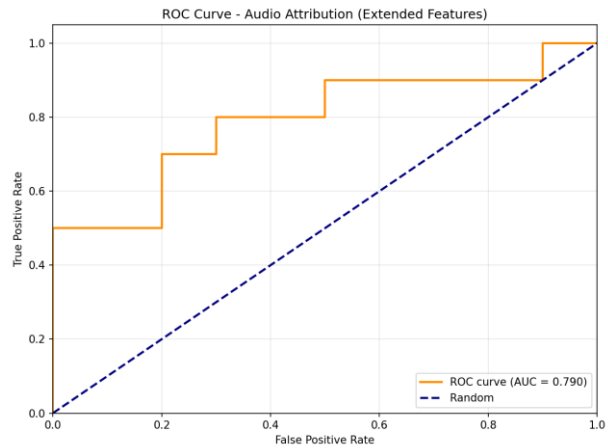
aggregation are manually defined and the resulting score is compared against a predefined threshold to determine whether two tracks are considered related.

In order to evaluate the initial system, a set of positive pairs and negative pairs (unrelated) is selected from the MIPPIA/SMP dataset. Performance is measured using ROC curves and AUC, along with summary statistics such as average similarity and false positive rates.

While limited by its reliance on handcrafted features and fixed weighting, it establishes a foundation for understanding the attribution task and guiding more advanced, learned approaches.

Some indicative results using the MIPPIA/SMP dataset with a threshold of 0.7 can be found below:





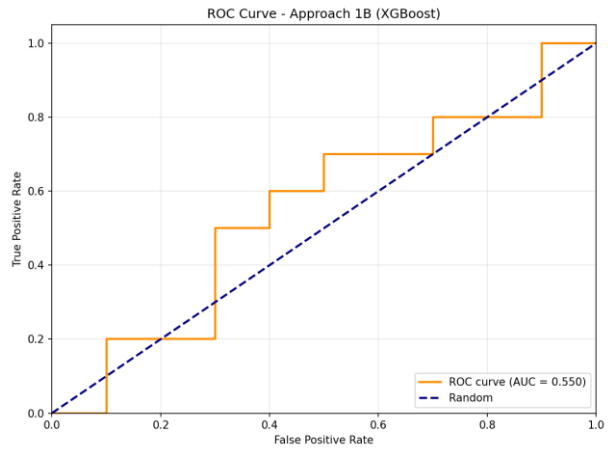
Considering all songs are real, the standard features seem to be able to capture most characteristics and so extended features don't improve significantly and also tend to be more inconsistent. Through the CLI several combinations can be given as input, however since this is a heuristic approach and the weights are manually defined the results can be inconsistent.

Moving to the second approach, in order to move past the heuristic elements of the baseline, the attribution task is formulated as a data-driven pairwise similarity modeling problem though XGBoost. Instead of manually combining similarity metrics, a model is trained to learn how to map similarity features between two tracks into a single relational continuous score.

The preprocessing, feature extraction and similarity metrics calculation remain identical to the baseline approach. Instead of combining similarity metrics using fixed weights, an XGBoost model is trained to learn a mapping from similarity features to a continuous attribution score. This allows the model to capture nonlinear interactions between features and adaptively weight their importance based on observed patterns in the training data. The model is trained using positive and negative pairs from MIPPIA/SMP dataset. Specifically, the training setup consists of:

- Training set: 50 positive and 50 negative pairs
- Validation set: 10 remaining pairs
- Hyperparameters were chosen as: max_depth=3 and n_estimators=100
- Predict_proba was chosen as similarity score

The XGBoost model achieved ROC AUC 0.55 on the validation set, compared to ~0.78 for the heuristic baseline. This counter-intuitive result highlights a limitation, where with only 50 training pairs, the model severely overfit to training data patterns and did not generalize. Training accuracy exceeded 95%, but the learned feature weights performed worse than the manually designed baseline on held-out data. Given time constraints, extensive hyperparameter tuning and cross-validation were not pursued.




```
============================================================
Training XGBoost model on MIPPIA
============================================================

Extended features: ENABLED
Pairs available: 50

Processing positive pairs...
  10/50
  20/50
  30/50
  40/50
  50/50
Generating negative pairs...
  10/50
  20/50
  30/50
  40/50
  50/50

Training XGBoost...
  Samples: 100 (positive=50, negative=50)
  Features: ['mfcc', 'chroma', 'spectral', 'tempo', 'dtw', 'phase_continuity', 'hnr', 'spectral_flux']

  Train accuracy: 1.000
  Test accuracy: 0.650
  ROC AUC: 0.530

============================================================
Feature Importances:
  mfcc               : 0.132
  chroma             : 0.124
  spectral           : 0.121
  tempo              : 0.115
  dtw                : 0.181
  phase_continuity   : 0.116
  hnr                : 0.129
  spectral_flux      : 0.081
============================================================

Model saved to: model.pkl
```

Below are some more detailed examples:

```
Processing track A: smp_dataset/final_dataset_clean/3/The Gap Band - Oops Upside Your Head.wav
Processing track B: sonics_dataset/fake_songs/fake_songs/fake_00001_suno_0.mp3

============================================================
Similarity Score (XGBoost): 0.314
Attribution: NO
============================================================
```

```
ataset/final_dataset_clean/5/Jennifer Rush - The Power Of Love _Official Video_ _VOD_.wav" --model model.pkl
✓ Loaded model from: model.pkl
Processing track A: smp_dataset/final_dataset_clean/1/Summer Dream.wav
Processing track B: smp_dataset/final_dataset_clean/5/Jennifer Rush - The Power Of Love _Official Video_ _VOD_.wav

============================================================
Similarity Score (XGBoost): 0.070
Attribution: NO
============================================================
```

```
✓ Loaded model from: model.pkl
Processing track A: smp_dataset/final_dataset_clean/3/The Gap Band - Oops Upside Your Head.wav
Processing track B: smp_dataset/final_dataset_clean/3/Mark Ronson - Uptown Funk _Official Video_ ft_ Bruno Mars.wav

============================================================
Similarity Score (XGBoost): 0.961
Attribution: YES
============================================================
```

Limitations and Future Steps

Even though these approaches serve as a baseline to explore the attribution task, the generalization ability is limited. This work was partially constrained by the limited training data of MIPPIA/SMP and enhancing with the full catalog of SONICS dataset could be useful. Additionally, the feature set could be enhanced with feature embeddings from pretrained models, an approach that was explored but did not work under the current setup.

Advanced architectures, such as Siamese networks with triplet or contrastive loss could be explored or attention based mechanisms could be used to learn similarities directly from audio.